SANGER: CLONED AMPLICONS

one clone is one molecule, read once



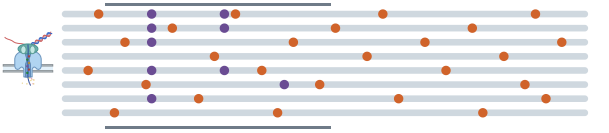
each clone's own mutations, exactly

529-bp window that the clones read cleanly;
447 chromatograms from the six mice of Figure 5

● mutation ● sequencing error

NANOPORE: CLONING-FREE AMPLICONS

every read spans the 1,230-bp amplicon



500 reads per library. A mutation is called where more reads carry it than the controls explain; one seen in a single read is only estimated.

Only from Sanger clones

Each molecule's own mutations
the mutation load of every clone, the share of clones mutated, and frequency as published: unique mutations over the bases each clone read

Bookkeeping checks
copied traces, clones filed under two mice, mutated clones filed as controls, mislabeled tubes (Figure 5E)

Error reported, not subtracted
the AID-null clones' error rate (about 20 per 10^5 bases at Q ≥ 20 in both published studies), and how many of each animal's calls it explains

From both, and they agree

Wild type against AID-null
Sanger 38-fold; nanopore at least 141-fold

Substitution spectrum
r = 0.92 (95% CI 0.74–0.98), slope 0.97

Position along the window
r = 0.94 in 25-bp bins

AID hotspot use
G:C at WRC/GYW 58% and 54% (20% of sites); A:T at WA/TW 70% and 69% (50% of sites)

Not comparable: absolute density
a clone is read once; nanopore counts unique mutations among 500 reads. Compare densities within a platform, not between.

Only from nanopore reads

The whole amplicon
positions 90–1,230: 229-fold over controls; 69% of mutations in the first 450 bp, falling 7-fold to the last 250 bp (Figure 6A)

Clones and mutation trees
clones of each mouse, and their mutations ordered into trees (Figures 6 and S10)

Calls at a known false-discovery rate
each candidate tested against the controls, the threshold set by them as decoys; single-read mutations as one background-subtracted estimate